SOLAX POWER

LV electrical distribution system and associated inverter type

ENGLISH

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In this article the following topics are included:

1. Earthing System

According to standard IEC 60364-3, the different grid earthing systems are defined as: TN, TT & IT systems; and the TN system is further separated into TN-C, TN-S, TN-C-S.

- 'T' means direct connection to the earth;
- 'I' means isolated from earth, connected to earth through an impedance;
- 'N' means direct connection to the earthed point(Neutral);
- 'S' – Protective function is provided by a conductor separate from Neutral;
- 'C' – Neutral and protective functions are combined in PEN conductor;

Thus you can understand the TN, TT & IT systems in the following notes:

- ➢ TT: transformer neutral earthed and frame earthed.
- ➢ TN: transformer neutral earthed, frame connected to neutral.
- ➢ IT: unearthed transformer neutral, earthed frame.

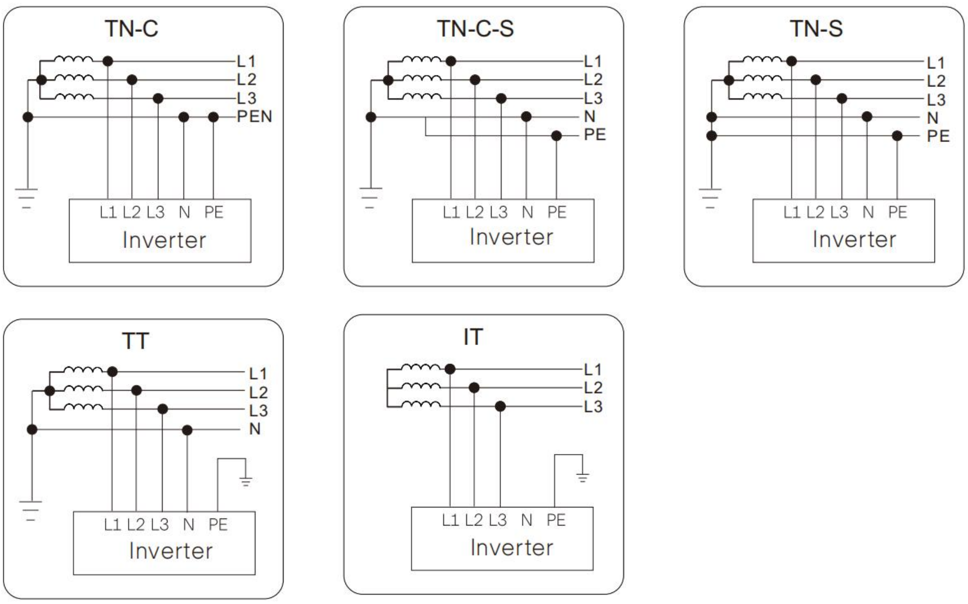


Fig. 1

Next, we will give brief introduction of the grid systems in different global regions.

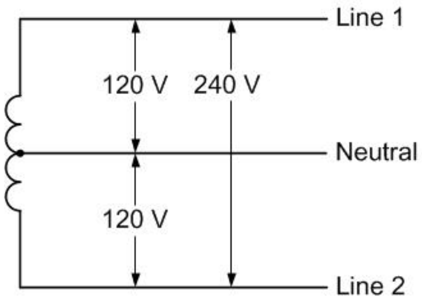
2. North American Grid System

According to ANSI C84-la-1980, there are two different LV distribution grid types:

① Split Phase - 120/240V

② Delta - 208V&240V and Wye - 277/480V, 120/208V

The most common residential service in North America is split phase 120/240V, which are used to power 120V lighting and plug loads and power 240V single phase loads such as a water heater, electric range, or air conditioner.

Split Phase, 120/240 V	
L-L: 240V L-N: 120V	 Fig. 2
Voltage	*Applicable Type*
L-L/L-N 240V/120V	A1-Hybrid (3.8 ~ 7.6kW) A1-HYD-G2 (3.8 ~ 7.6kW) A1-AC-G2 (3.8 ~ 7.6kW) A1-Smart (3.8 ~ 7.6kW)

For commercial building the most common electric service is 120/208V Wye, which is used to power 120V plug loads, lighting, and smaller HVAC systems. In larger facilities the voltage is 277/480V and used to power single phase 277 volt lighting and larger HVAC loads.

3-Phase, Wye 120/208 V, 277/480V

L-L: 208V

L-N: 120V

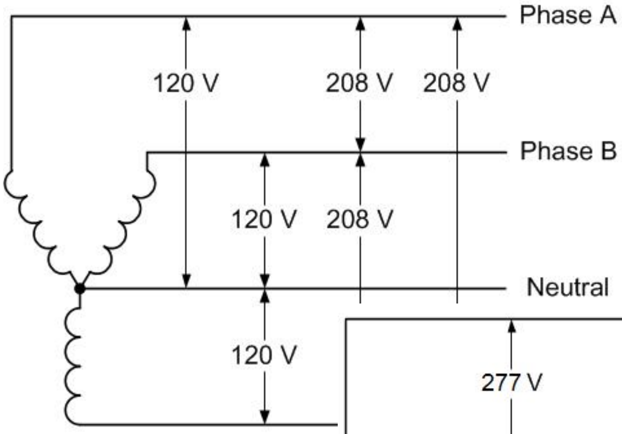


Fig. 3

L-L: 480V

L-N: 277V

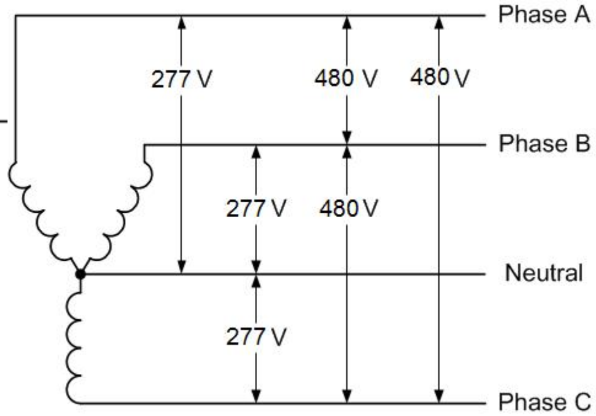
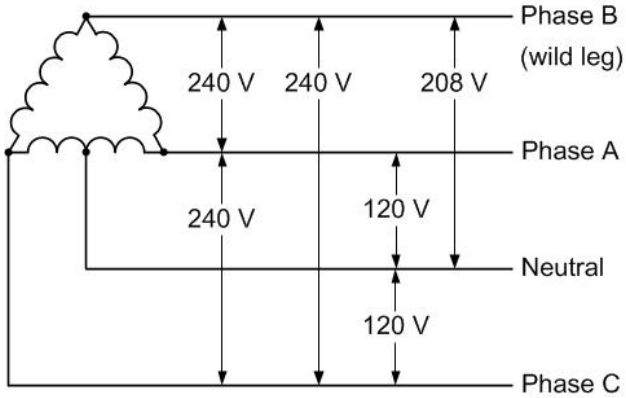


Fig. 4

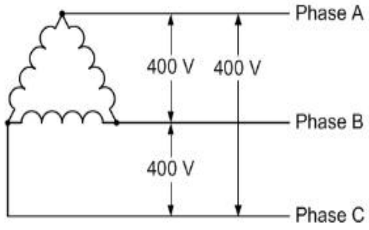
The following grid type is also known as a high-leg or wild-leg delta system. Usually used in older manufacturing facilities with mostly three-phase motor loads and some 120 volt single-phase lighting and plug loads.

3-Phase, Delta 208, 240V/120V Wild Leg	
L-L 240V L-N 208V 120V(Split)	 Fig. 5

3. EU Grid System

In Europe, IT, TT, TN-C, TN-S, TN-C-S systems are commonly seen. The diagrams below will show the different LV distribution systems in Europe.

3-Phase, Wye 220/380V, 230/400V, 240/415V



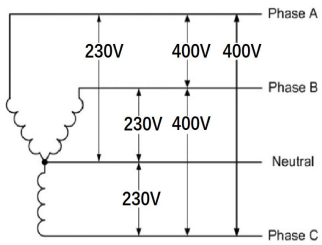


Fig.6

Voltage	Applicable Type
Three Phase L-L 380V/400V/415V	X3-MIC G2 (3~15kW) X3-PRO G2 (8~30kW) X3-MEGA-G2 (40~60kW) X3-FORTH (75/80/100/110/120/125kW) X3-HYBRID G4 (5~15kW) X3-FIT G4 (5~15kW)

Single Phase L-N 220V/230V/240V	X1-MINI (0.6~3.3kW) X1-BOOST (2.5~6kW) X1-Smart (6/7/8kW) X1-HYBRID G4 (3.7~7.5kW) X1-FIT G4 (3.7~7.5kW) X1-AC (3~5kW)

3-Phase, Delta 400 V (no neutral)	
Three Phase L-L 400V	Fig.7

Voltage	*Applicable Type*
Three Phase L-L 380V/400V/415V	X3-PRO G2 (8~30kW) X3-MEGA-G2 (40~60kW) X3-FORTH (75/80/100/110/120/125kW)

4. Latin American Grid System

The commonly supplied power is 3-Phase Wye 220/380V same as Fig. 6. However there are some other special types of grid supply which are also different in Latin America.

For example, in some regions of Brazil, there is no standard voltage. The nominal voltage of 3-Phase Wye can be 127/220V, 125/216V, 220/380V and 230/400V as below:

3-Phase, Wye 127/220V, 125/216V, 220/380V and 230/400V

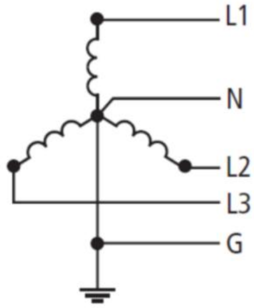
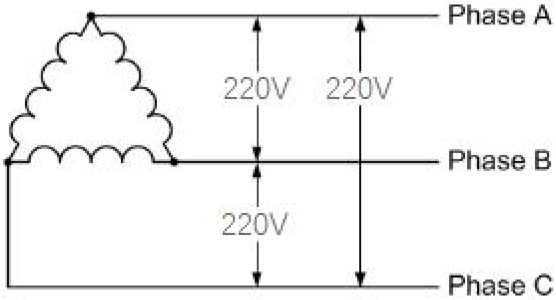


Fig.8

Voltage	Applicable Type
Three Phase L-L 380V/400V/415V	X3-MIC G2 (3~15kW) X3-PRO G2 (8~30kW) X3-MEGA-G2 (40~60kW) X3-FORTH (75/80/100/110/120/125kW) X3-HYBRID G4 (5~15kW) X3-FIT G4 (5~15kW)
Single Phase L-N 220V/230V/240V	X1-MINI (0.6~3.3kW) X1-BOOST (2.5~6kW) X1-Smart (6/7/8kW) X1-HYBRID G4 (3.7~7.5kW) X1-FIT G4 (3.7~7.5kW)

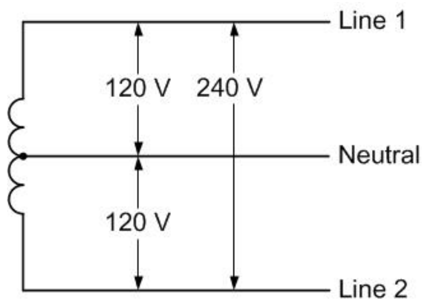
Besides that, most federative units use 220 V electricity (three-phase) as below:

3-Phase, Delta 220 V (LV Grid)

L-L 220V	 Fig.9

Voltage	*Applicable Type*
Three Phase L-L 220V	X3-MIC-G2-LV (5~8kW) X3-PRO-G2-LV (10~15kW) X3-MGA-G2-LV (20~30kW) X3-FTH-LV (40~70kW)

In Mexico, the residential households are using 2 phase 3 wire grid connection, also known as Split Phase 110V/220V or 120V/240V, similar to Fig. 2 in North American Market. We have the following compatible inverter specifically for Mexican market.

Split Phase, *110V/220V,* 120/240 V	
L-L: 220V 240V L-N: 110V 120V	 Fig. 10

Voltage	*Applicable Type*
L-L: 220V L-N: 120/127V	X1-SPT (3~7kW)

5. Grid System in Japan

Similarly to North American Split Phase grid, Japan has the 100/200V split phase system for normal household power supply.

1-Phase, 3-Wire 100/200V 50Hz(East Japan) 1-Phase, 3-Wire 100/200V, 105/210V 60Hz(West Japan)

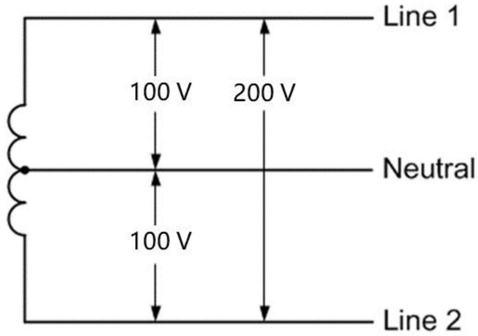


Fig. 11

Voltage	Applicable Type
L-N 100/105V L-L 200/210V	J1ESS-HB58/58-1 J1ESS-HB58X/115/173

3-Phase, Delta 200 V

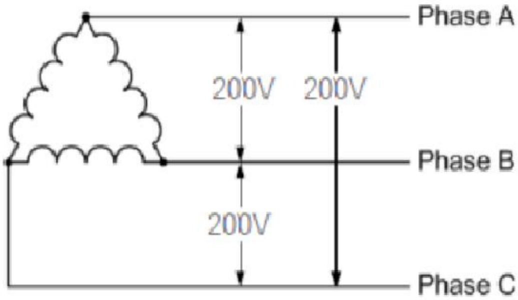


Fig. 12

Voltage	Applicable Type
L-L 200V	J1ESS-HB58/58-1 J1ESS-HB58X/115/173